

Subject: Digital Design
Dimitar Trajkovski

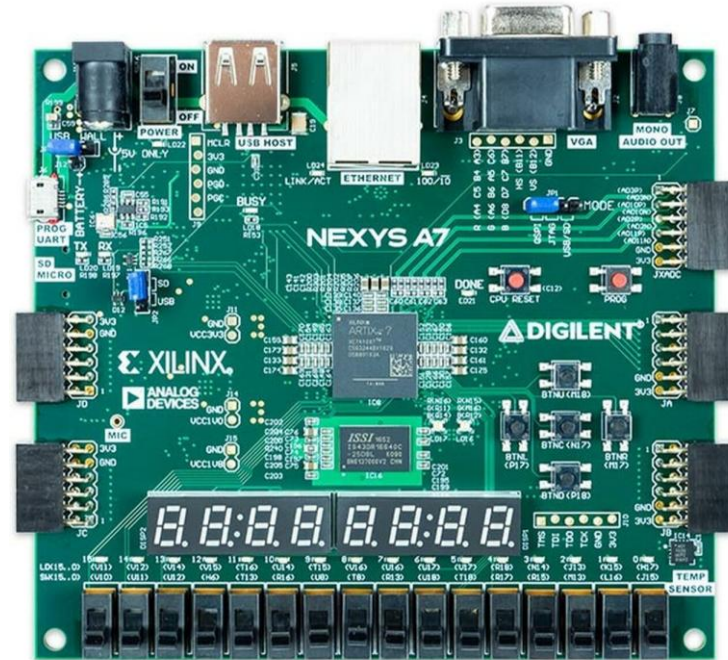


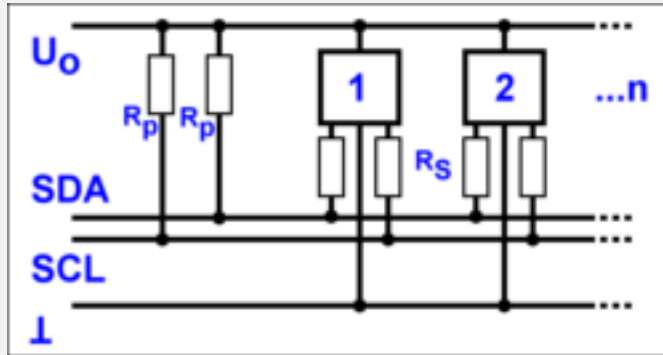
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01 I2C-Inter-
Integrated Circuit

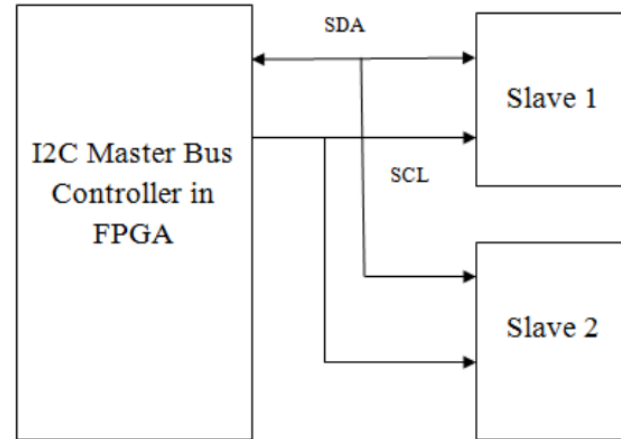
02 I2C Finite State
Machine

03 ADT7420 Temperature
sensor

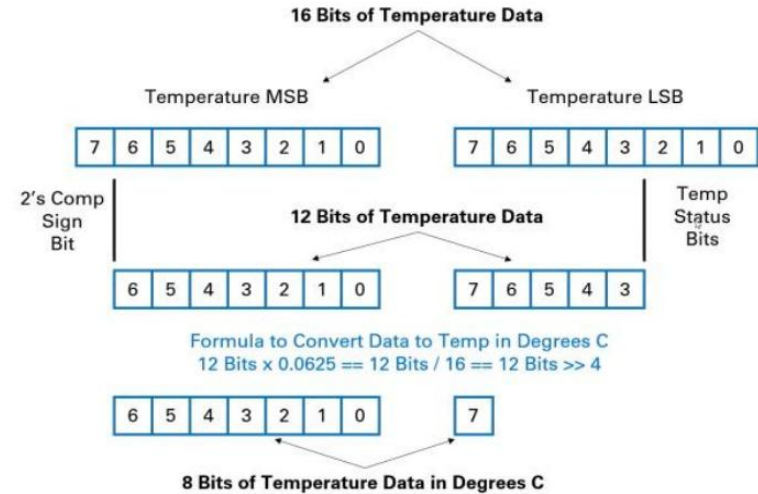
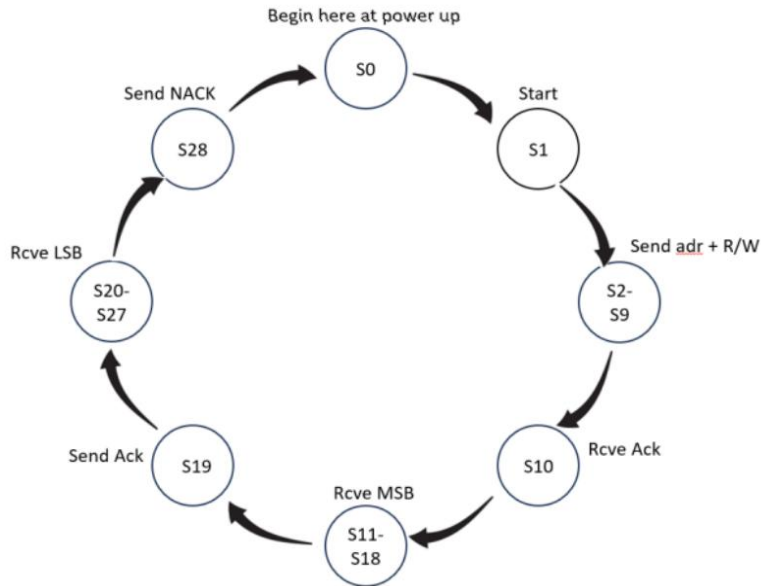
01 I2C



- SDA line
- SCL line



02 FSM



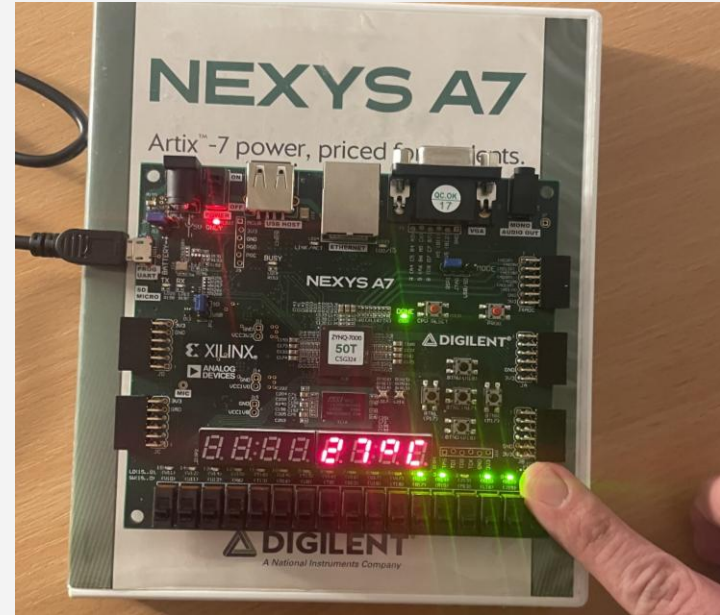
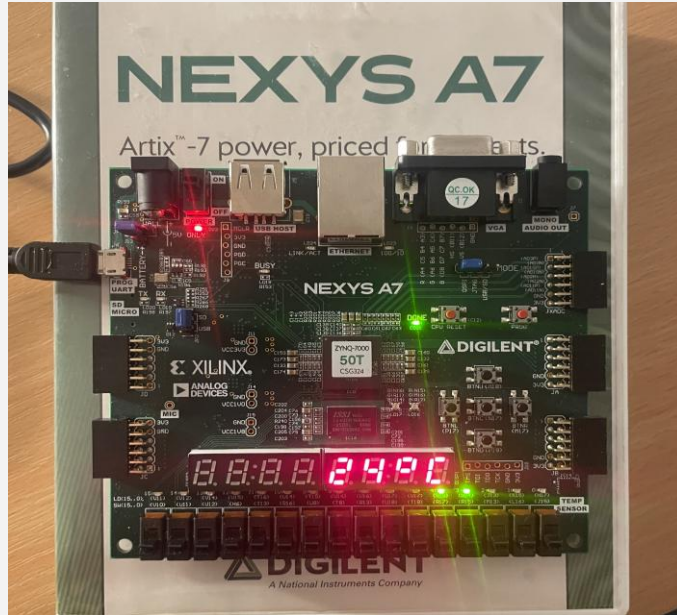
03 ADT7420

ADT7420 Register Map

Understanding the register structure is crucial for successful communication:

Register	Address	Function	Size
Temperature	0x00:0x01	Temperature value (MSB:LSB)	16-bit
Status	0x02	Device status flags	8-bit
Configuration	0x03	Operating mode settings	8-bit
T_HIGH	0x04:0x05	High temperature threshold	16-bit
T_LOW	0x06:0x07	Low temperature threshold	16-bit
T_CRIT	0x08:0x09	Critical temperature threshold	16-bit
ID	0x0B	Manufacturer ID (0xCB)	8-bit

The ADT7420 is renowned for its high precision, offering an accuracy of $\pm 0.25^{\circ}\text{C}$ and supporting up to 16-bit resolution.



Implementation